MAT 3007 – Optimization

Introduction to Optimization

Lecture 01

Instructor: Junchi Yang

CUHK-Shenzhen

Logistics

Meeting Time

MAT 3007 – Optimization (L-03)

- Location: TxC301
- Lecture Session 01: Monday, 3:30 p.m. – 4:50 p.m.
- Lecture Session 02: Wednesday, 3:30 p.m. – 4:50 p.m.

Contact Information

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- **Office Hours:** Wednesday, 11:00 a.m. – 12:00 p.m.

Teaching Assistants

TAs – Contact Information

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  Office Hours: Wed, 8 - 9 p.m.
- Jiaming Chen
  Email: 121090033@link.cuhk.edu.cn.

Tutorial Time

Tutorials:

- T-1: Mon 7:00 p.m. - 7:50 p.m. Teaching C Bldg 414.
- T-2: Mon 8:00 p.m. - 8:50 p.m. Teaching C Bldg 414.
- T-3: Wed 6:00 p.m. - 6:50 p.m. Teaching C Bldg 414.
- T-4: Wed 7:00 p.m. - 7:50 p.m. Teaching A Bldg 302.
- T-5: Thur 6:00 p.m. - 6:50 p.m. Teaching C Bldg 308.

- Tutorials will start in the second week!

Literature and Textbooks

Lecture slides will be provided.

Reference Books:

- Bertsimas, Tsitsiklis: *Introduction to Linear Optimization*, Athena Scientific, 1997.
- Boyd, Vandenberghe: *Convex Optimization*, Cambridge University Press, 2004.
- Beck: *Introduction to Nonlinear Optimization. Theory, Algorithms, and Applications with MATLAB*, MOS-SIAM Series Optimization, 2014.

Grading Scheme

- Assignments: **20%**.
  - There will be 9 assignments this semester. The lowest score will be dropped. Late submissions will not be accepted.
  - **Work Submission Guidelines**: Plagiarism is strictly prohibited. Discussion with peers is allowed, but each student must submit their own work. Submissions are to be made electronically.
- Midterm Exam: **40%**.
  - Tentative Schedule: Saturday, March 22nd, 10:00 - 11:30. Students are allowed to bring one self-prepared A4 cheat sheet (notes on both sides).
- Final Exam: **40%**.
  - Tentative Schedule: TBD.

What is Optimization?

Example : Maximum Area

Example 1: Maximum Area Problem

You have 80 meters of fencing and want to enclose a rectangular shape (for example a yard) – as large as possible (in area).

- How should you do it?

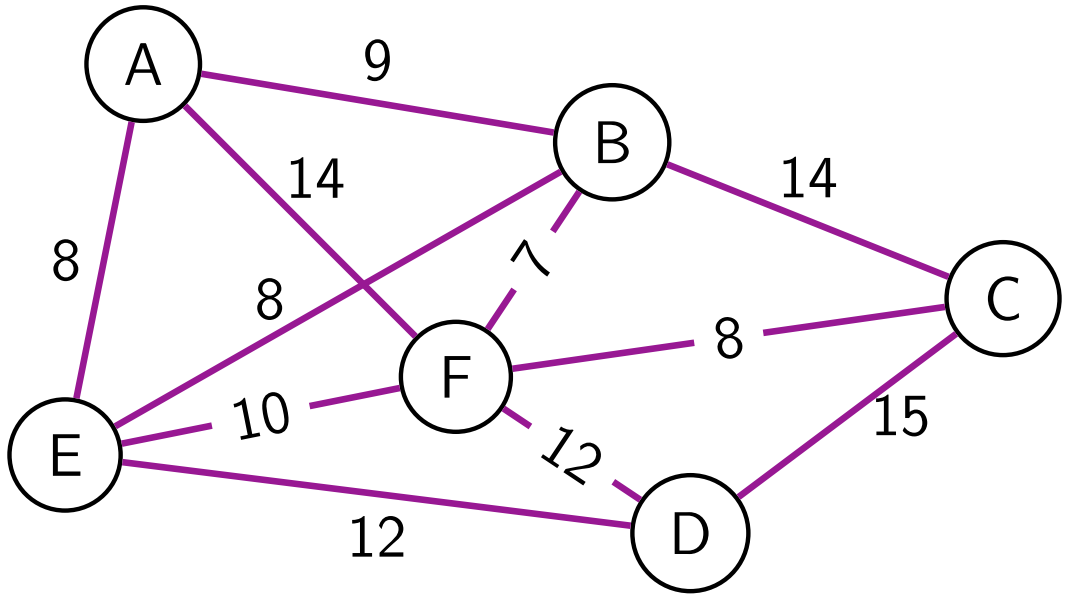




Example 2: Traveling Salesman Problem

A salesman needs to visit a number of places in a day.

- How should he schedule his trip so that the total distance is shortest (or the total cost is smallest)?



Example 3: Production Problem

A company produces two types of alloy products. They need to use steel, iron and copper as the resource. The production requirement and profit for each type of alloy are given as follows:

	Steel	Iron	Copper	Profit
Alloy 1	1	0	1	\$1
Alloy 2	0	2	1	\$2
Resources	100	200	150	

- How much of each type of products should the company produce in order to maximize its profit?

Optimization

Key Components of Optimization Problems:

- Decision.
- Objective.
- Constraints.

- Optimization concerns choosing a decision (or decisions) to optimize certain objectives while subject to certain constraints.

- Optimize could mean maximize or minimize depending on the problem context.

Back to Our Examples

Maximum Area Problem:

- Decision: length and width of the yard; Objective: maximize the area; Constraints: the total length of the fence.

Traveling Salesman Problem:

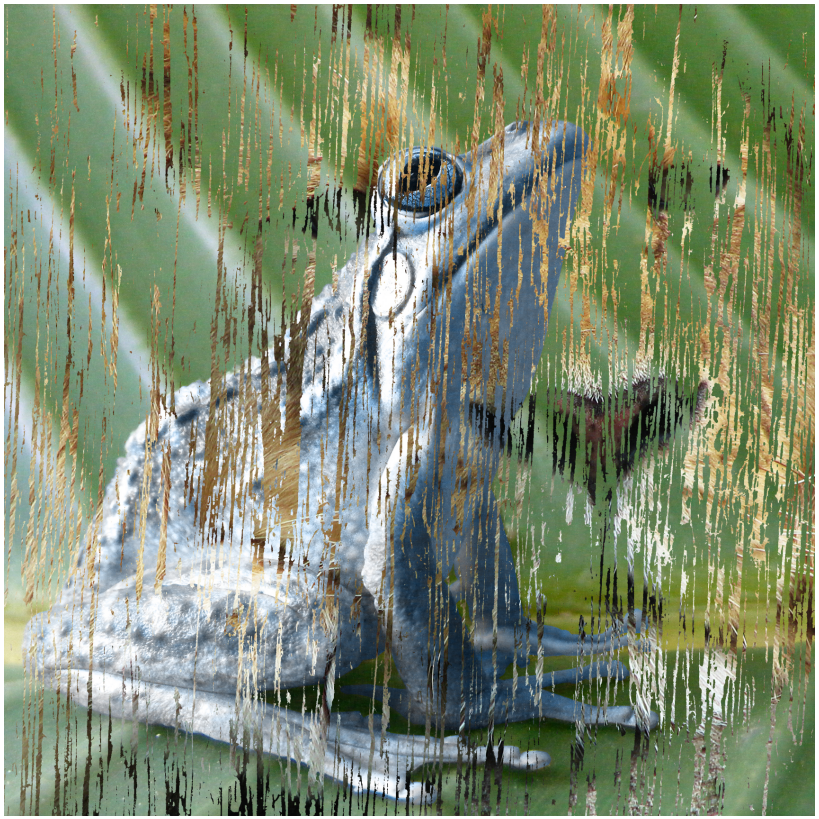
- Decision: the order of visits; Objective: minimize the total distance; Constraints: has to visit each place once.

Production Problem:

- Decision: production plan; Objective: maximize profit; Constraints: resource.

Example 4: Removing Scratches





- How can we remove the gray scratches while keeping the undamaged parts of the image?
  - Find the missing pixels.

Example 4: Removing Scratches





- A possible "nicer" reconstruction.

Example 4: Removing Scratches





- Decision: the whole image;
- Objective: have a "nicer" reconstructed image;
- Constraints: do not change undamaged parts.

Optimization

Optimization is ubiquitous in our life:

- Economics, finance, science, engineering, business, government, sports, marketing, etc.

Nothing at all takes place in the Universe in which some rule of maximum or minimum does not appear. – L. Euler, 1707-1783



Content of the Course

Modeling Techniques:

How to translate a practical problem into an optimization problem, especially a good optimization problem?

Optimization Algorithms:

How to solve optimization problems efficiently?

- Some fundamental algorithms (simplex, gradient, Newton, interior point methods) will be discussed.

Implementation:

How to solve optimization problems using available tools?

- Computer software will be used: MATLAB (Python, etc).

Basic Terminologies

Optimization Problems

Mathematically, an optimization problem is usually represented as:

Optimization Problem:

$$\begin{aligned} \text{maximize/minimize}_x \quad & f(x) \\ \text{subject to} \quad & x \in \Omega. \end{aligned} \tag{1}$$

Setup:

- x is the decision or optimization variable.
- $f : \mathbb{R}^n \to \mathbb{R}$ or $f : \Omega \to \mathbb{R}$ denotes the objective function.
- $\Omega \subset \mathbb{R}^n$ is the feasible set $\rightsquigarrow$ constraints.

In practice, we often consider a slightly more restrictive format:

Optimization Problem:

$$\begin{aligned} \text{minimize}_{x\in\mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \le 0, \quad \forall\ i = 1, \ldots, m, \\ & h_j(x) = 0, \quad \forall\ j = 1, \ldots, p. \end{aligned}$$

- Here, $g : \mathbb{R}^n \to \mathbb{R}^m$ and $h : \mathbb{R}^n \to \mathbb{R}^p$ model inequality and equality constraints:

$$g(x) = \begin{bmatrix} g_1(x) \\ \vdots \\ g_m(x) \end{bmatrix} \in \mathbb{R}^m \quad \text{and} \quad h(x) = \begin{bmatrix} h_1(x) \\ \vdots \\ h_p(x) \end{bmatrix} \in \mathbb{R}^p.$$

In practice, we often consider a slightly more restrictive format:

Optimization Problem:

$$\begin{aligned} \text{minimize}_{x\in\mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \le 0, \quad \forall\ i = 1, \ldots, m, \\ & h_j(x) = 0, \quad \forall\ j = 1, \ldots, p. \end{aligned}$$

We can express this problem using the abstract format:

$$\begin{aligned} \Omega &:= \{x \in \mathbb{R}^n : g_i(x) \le 0,\ i = 1, \ldots, m,\ h_j(x) = 0,\ j = 1, \ldots, p\} \\ &= \{x \in \mathbb{R}^n : g(x) \le 0,\ h(x) = 0\}. \end{aligned}$$

Basic Terminologies

- Feasible point/Feasible solution: A decision x that satisfies all constraints.
- Domain/Feasible set/Feasible region Ω: The set of all feasible points.

- Optimal solution: A (feasible) decision variable that attains an objective value that is as good as any other feasible point.
- Optimal value: The infimum of f over the feasible set. If an optimal solution x^* exists, it is simply $f(x^*)$.

- In the case $\Omega = \mathbb{R}^n$ or $m = p = 0$, we say that problem (1) is unconstrained. Otherwise, (1) is a constrained optimization problem.

Example: A Simple Optimization Problem

$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ decision variable

objective

$$\begin{aligned} \text{minimize}_{x \in \mathbb{R}^2} \quad & f(x) := x_1 + 2x_2 \\ \text{subject to} \quad & x_1 + x_2 = 1 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$h(x) = x_1 + x_2 - 1$

$g_1(x) = -x_1$

$g_2(x) = -x_2$

$$\begin{aligned} \min_x \ & f(x) \\ \text{s.t} \ & h(x) = 0 \\ & g(x) \leq 0 \end{aligned}$$

Some Facts about Optimization Problems:

- The feasible set can be empty (the problem is said to be infeasible or degenerate in this case).

$$\min_x f(x) \quad \text{s.t.} \quad x \geq 0, \; x \leq -1$$

Some Facts about Optimization Problems:

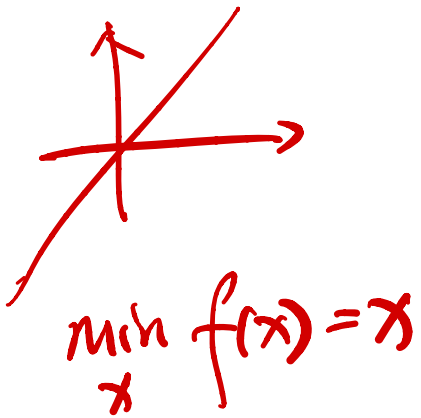
- The feasible set can be empty (the problem is said to be infeasible or degenerate in this case).
- An optimization problem may not always have an optimal solution.



$$\min_x \ f(x) = \frac{1}{x}$$
$$\text{s.t.} \quad x \geq 1.$$

Some Facts about Optimization Problems:

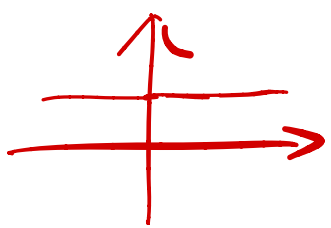
- The feasible set can be empty (the problem is said to be infeasible or degenerate in this case).
- An optimization problem may not always have an optimal solution.
- The optimal value can be unbounded.



Properties of Optimization Problems

Some Facts about Optimization Problems:

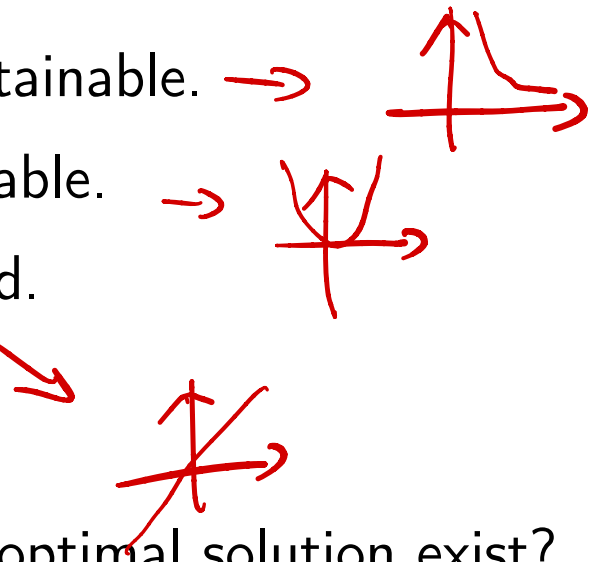
- The feasible set can be empty (the problem is said to be infeasible or degenerate in this case).
- An optimization problem may not always have an optimal solution.
- The optimal value can be unbounded.
- Even if an optimization problem is feasible, finite, and can attain its optimal value, the optimal solution may not be unique. (However, the optimal value must be unique).



Possible States for Optimization Problems

By classifying the different outcomes, an optimization problem may have the following states:

1. Infeasible.
2. Feasible, optimal value finite but not attainable.
3. Feasible, optimal value finite and attainable.
4. Feasible, but optimal value is unbounded.



Key Questions:

- When is the problem feasible? Does an optimal solution exist?
- How to determine if a candidate x is an optimal solution?
- How can we find optimal solutions numerically and efficiently?

Basic Terminologies and Formal Definitions

Background and Notation

- In the following, we briefly repeat some basic mathematical concepts and definitions.
- We then give a formal definition of minimizer and maximizer.

$\rightsquigarrow$ These are the "optimal solutions" we seek.

Vectors and Norms

Vectors and Norms

A point or vector $x \in \mathbb{R}^n$ is represented by a column vector

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \text{with components} \quad x_i \in \mathbb{R}, \quad \forall\, i.$$

The transposed $x^\top$ is the row vector $x^\top = (x_1, x_2, \cdots, x_n)$.

For $x, y \in \mathbb{R}^n$, we define the Euclidean inner product by

$$\langle x, y \rangle := \sum_{i=1}^n x_i y_i = x^\top y.$$

The inner product has the following properties:

- Positivity: $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ if and only if $x = 0$,
- Symmetry: $\langle x, y \rangle = \langle y, x \rangle$,
- Linearity: $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$,

for all $\alpha, \beta \in \mathbb{R}$ and $x, y, z \in \mathbb{R}^n$.

Vectors and Norms - II

The Euclidean norm of a point $x \in \mathbb{R}^n$ is defined as:

$$\|x\|_2 = \|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x^\top x} = \sqrt{x_1^2 + x_2^2 + ... + x_n^2}.$$

The Euclidean norm satisfies the following properties for all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

- Positivity: $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = 0$.
- Homogeneity: $\|\alpha x\| = |\alpha| \|x\|$.
- Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$.
- Cauchy-Schwarz inequality: $|\langle x, y \rangle| \leq \|x\| \|y\|$.

Other norms:

- ℓ_1-norm: $\|x\|_1$, ℓ_p-norm: $\|x\|_p = [\sum_{i=1}^n |x_i|^p]^{1/p}$, maximum norm $\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$, ...

Sets

Supremum and Infimum

The supremum of a nonempty set $\Omega \subset \mathbb{R}$ is the smallest scalar y such that

$$y \geq x \quad \text{for all} \quad x \in \Omega.$$

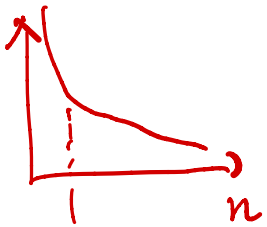
The infimum of a set $\Omega \subset \mathbb{R}$ is the largest scalar y such that

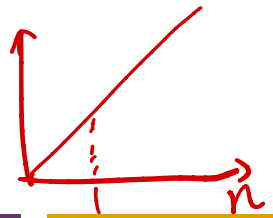
$$y \leq x \quad \text{for all} \quad x \in \Omega.$$

If $\sup \Omega \in \Omega$ ($\inf \Omega \in \Omega$) then $\sup \Omega = \max \Omega$ ($\inf \Omega = \min \Omega$).

Examples:

- $\sup\{1/n : n \geq 1\} = ?$, $\inf\{1/n : n \geq 1\} = ?$
- $\sup\{n : n \geq 1\} = ?$, $\inf\{n : n \geq 1\} = ?$





Local and Global Minimizer

Formal Definition: Minimizer

Let $\Omega \subseteq \mathbb{R}^n$ be a nonempty set and let $f : \Omega \to \mathbb{R}$ be given. For $\varepsilon > 0$ and $x \in \mathbb{R}^n$ we define the neighborhood

$$B_\varepsilon(x) = \{y \in \mathbb{R}^n : \|x - y\| < \varepsilon\}.$$

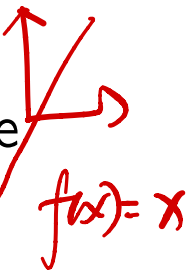
The point $x^* \in \mathbb{R}^n$ is said to be a:

- local minimizer of problem (1), if $x^* \in \Omega$ and there exists $\varepsilon > 0$ such that $f(x) \geq f(x^*)$ for all $x \in \Omega \cap B_\varepsilon(x^*)$.
- strict local minimizer of the problem (1), if $x^* \in \Omega$ and there is $\varepsilon > 0$ with $f(x) > f(x^*)$ for all $x \in (\Omega \cap B_\varepsilon(x^*))\backslash\{x^*\}$.

- global minimizer of (1), if $x^* \in \Omega$ and we have $f(x) \geq f(x^*)$ for all $x \in \Omega$.
- strict global minimizer of (1), if $x^* \in \Omega$ and it holds that $f(x) > f(x^*)$ for all $x \in \Omega\backslash\{x^*\}$.

Remarks and Comments

- global minimizer $\equiv$ global solution $\equiv$ optimal solution.
- We say the objective value is unbounded or the optimal value is $-\infty$, if for every $B \in \mathbb{R}$ there exists $x \in \Omega$ with $f(x) \leq B$.
- The definition for maximizer is identical, just change : $\geq / >$ $\to \leq / <$.



- The following optimization problems are equivalent:

$$\max_x \ f(x) \quad \text{s.t.} \quad x \in \Omega \quad \text{and} \quad \min_x -f(x) \quad \text{s.t.} \quad x \in \Omega.$$

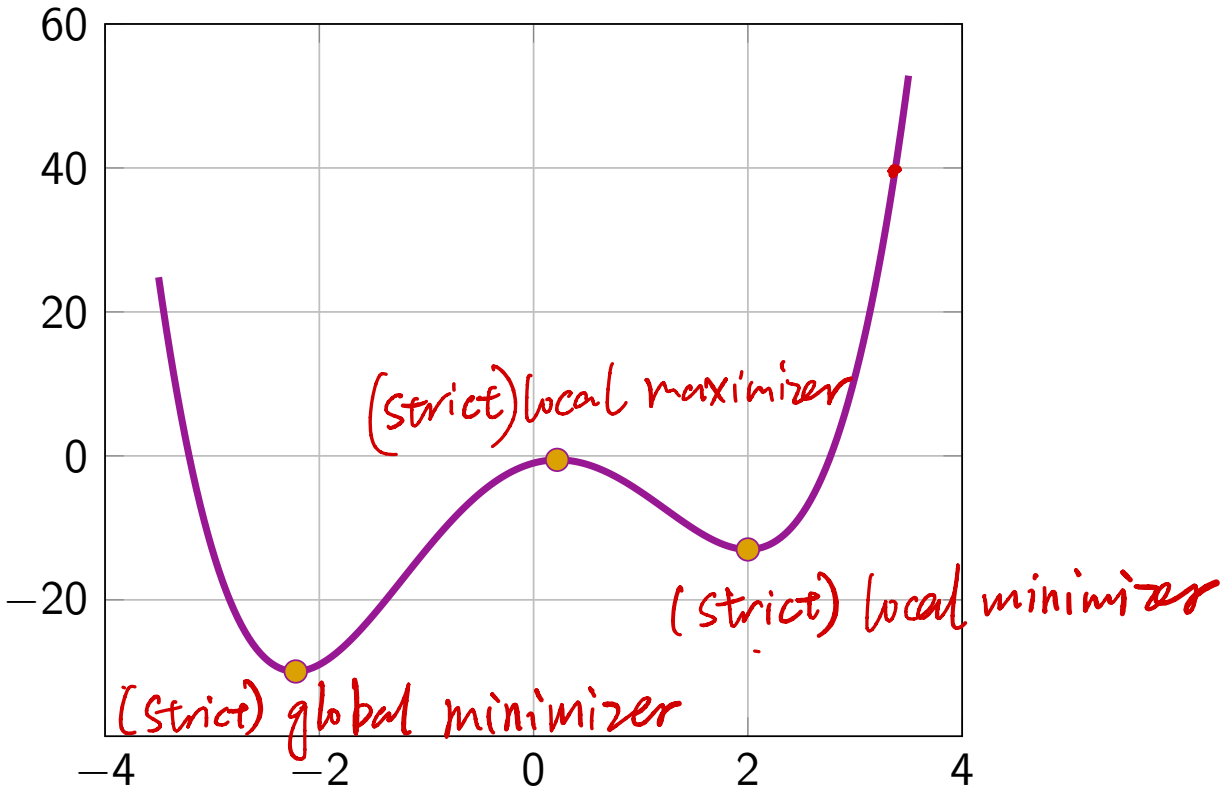
Every loc. maximum of f on Ω is a loc. minimum of $-f$ on Ω.

$\rightsquigarrow$ We will concentrate on minimization problems.

Example: Minimizer

We consider the unconstrained problem

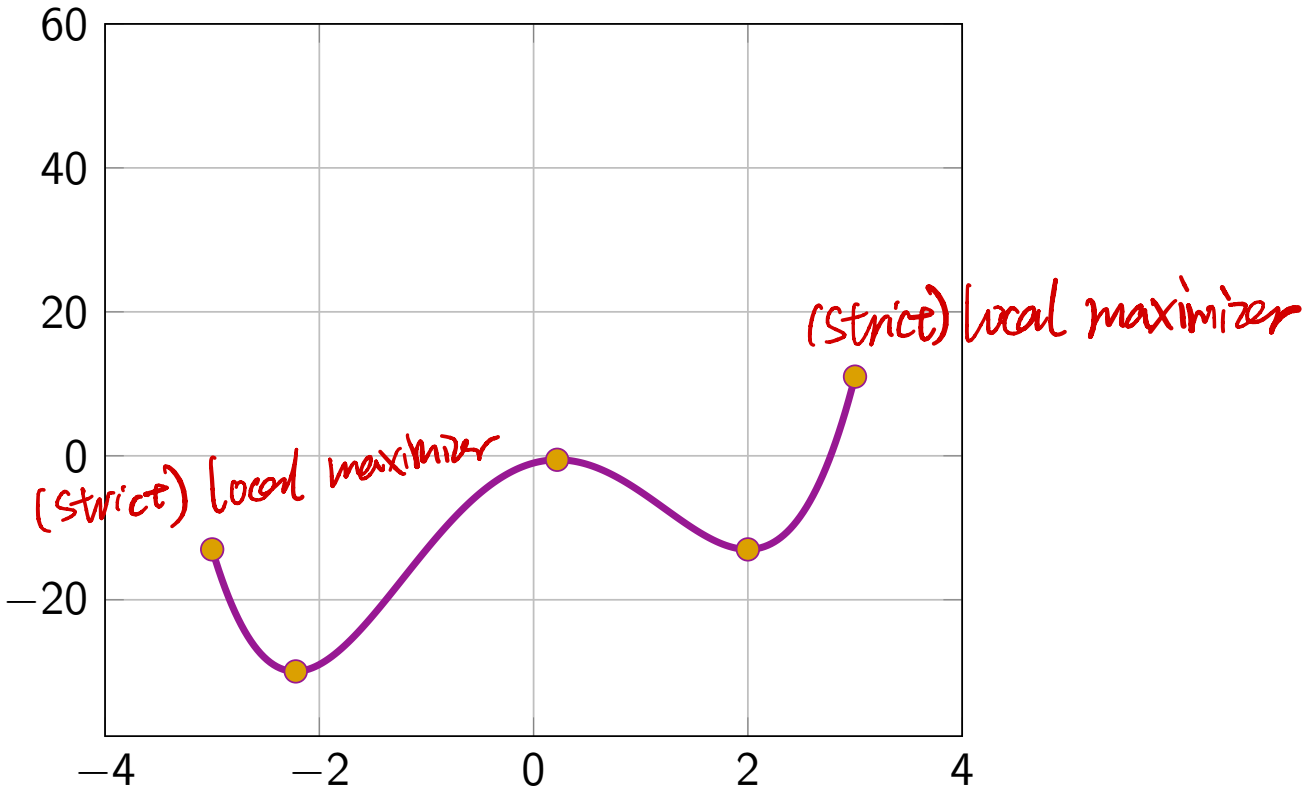
$$\text{minimize}_{x\in\mathbb{R}}\ f(x) := x^4 - 9x^2 + 4x - 1.$$



Example: Minimizer

Let us consider the constrained problem

$$\begin{aligned} \text{minimize}_{x\in\mathbb{R}} \quad & f(x) := x^4 - 9x^2 + 4x - 1. \\ \text{subject to} \quad & x \in [-3, 3]. \end{aligned}$$

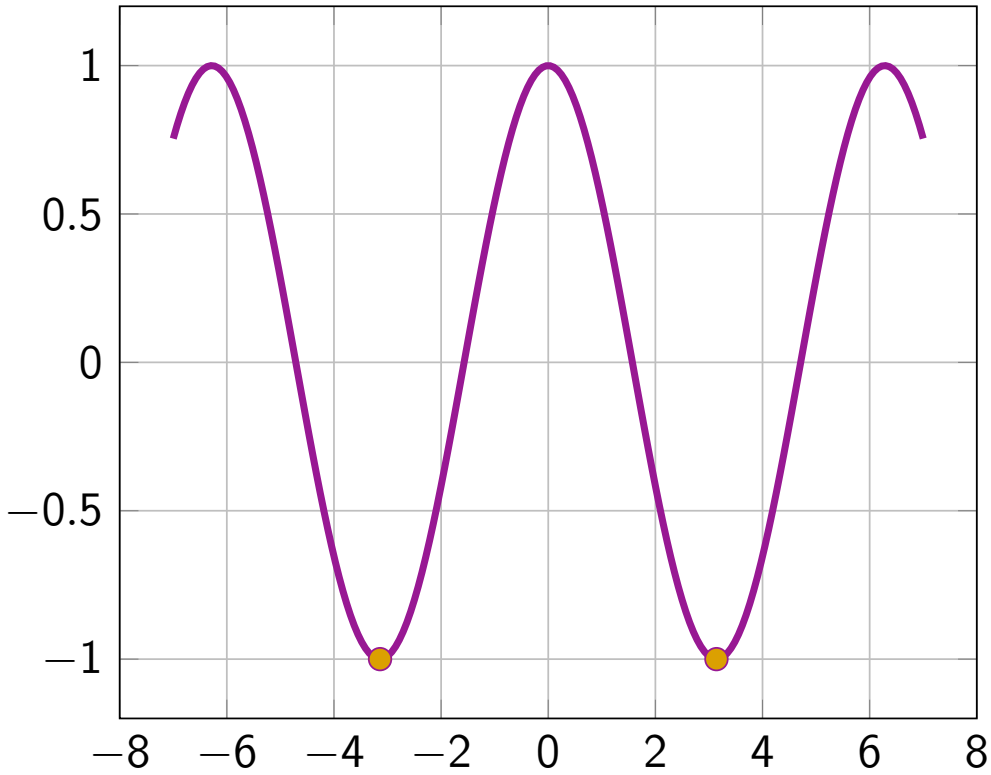


Example: Minimizer

We consider the unconstrained problem

$$\text{minimize}_{x \in \mathbb{R}} \; f(x) := \cos(x).$$

- All points in $S = \{(2k+1)\pi : \pi \in \mathbb{Z}\}$ are global minima.
- None of the global minima is strict.



Classification of Optimization Problems

Optimization Problem:

$$\begin{aligned} \text{minimize}_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad \forall\ i = 1, ..., m, \\ & h_j(x) = 0, \quad \forall\ j = 1, ..., p. \end{aligned}$$

- Unconstrained optimization: If $\Omega = \mathbb{R}^n$ or $m = p = 0$. Otherwise: constrained optimization.
- Linear optimization (LP): Constraints and objective are linear.
- Nonlinear optimization (NLP): Either some of the constraints or the objective function is nonlinear.
- Integer/Discrete optimization (IP): Some of the decision variables have to be integers ($x \in \mathbb{Z}^n$).
- Continuous optimization: All the decision variables can take continuous values.

More on Linear Functions

- A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is called linear if

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y), \quad \forall\, x, y \in \mathbb{R}^n, \quad \alpha, \beta \in \mathbb{R}.$$

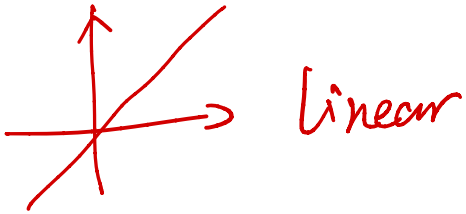
- A function $g : \mathbb{R}^n \to \mathbb{R}^m$ is called affine-linear if we can write $g(x) = f(x) - b$ for some linear function f and $b \in \mathbb{R}^m$.

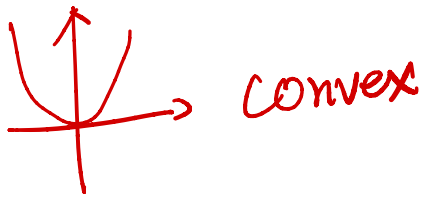
$\rightsquigarrow$ Typically $g(x) = Ax - b$ where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$.

$\rightsquigarrow$ Much more on linear problems in the next weeks!

Classifications

- Constrained vs Unconstrained.
- Linear vs Nonlinear.
- Convex vs Nonconvex.
- Continuous vs Discrete.





- By default, when we talk about an optimization problem, we assume it is a continuous problem, unless we explicitly state that it is discrete.

Remarks

The above classifications are based on the structure of f, g, h, and of the feasible set Ω:

- Sometimes an NLP can be equivalently transformed to an LP.
- Sometimes an IP can be equivalently transformed to a continuous optimization problem.

Linear optimization is the most well-studied and the easiest optimization problem:

- Nonlinear optimization and integer optimization could be significantly harder than LP.
- Therefore, in many cases, people strive to find LP formulations for problems.

Outline of the Lecture: Topics

- **Week 1:** Introduction to optimization.
- **Weeks 2 – 4:** Basics of linear optimization and the simplex method.
- **Weeks 4 – 5:** Duality theories for linear programs.
- **Weeks 6 – 7:** Dual simplex, sensitivity analysis, and advanced topics in linear programming.
- **Weeks 7 – 8:** Review and midterm exam.
- **Weeks 8 – 11:** Nonlinear optimization: Optimality conditions, convexity, and foundational concepts.
- **Weeks 11 – 13:** Algorithms for nonlinear problems.
- **Weeks 13 – 15:** Introduction to integer programming and basic algorithms.

Questions?